
HOMEWORK DAY 21 – Series §11.2

1. (a) What does it mean to state that $\sum_{k=1}^{\infty} a_k = 5$?

(b) Assume $\sum_{k=1}^{\infty} a_k = 5$. What is $\lim_{k \rightarrow \infty} a_k$?

What is $\lim_{n \rightarrow \infty} s_n$, where $s_n = \sum_{k=1}^n a_k$?

(c) Find the value of the series $\sum_{k=1}^{\infty} a_k$ if its partial sums are given by $s_n = \frac{n^2 - 1}{4n^2 + 1}$.

(d) Find $\sum_{n=1}^{\infty} \cos n$ or determine it does not converge.

(e) Find $\sum_{n=1}^{\infty} \cos(n\pi)$ or determine it does not converge.

2. Fill in the blanks using either *may* or *must*.

(a) A series $\sum_{n=1}^{\infty} a_n$ whose terms a_n tend to 0 as $n \rightarrow \infty$ _____ converge.

(b) A series $\sum_{n=1}^{\infty} a_n$ that converges _____ have terms a_n that tend to 0 as $n \rightarrow \infty$.

(c) If a series $\sum_{n=1}^{\infty} a_n$ diverges, then the terms a_n _____ not tend to 0 as $n \rightarrow \infty$.

(d) If a series diverges, then the Divergence Test _____ succeed in proving the divergence.

(e) If $\sum_{n=10}^{\infty} a_n$ diverges, then $\sum_{n=1000}^{\infty} a_n$ _____ diverge.

3. Evaluate the following series or determine that they diverge (with very brief explanation).

(a) $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$

(b) $\sum_{n=2}^{\infty} 0.4^n$

(c) $\sum_{n=-3}^{\infty} \frac{1}{(-5)^n}$

(d) $1 + 0.4 + 0.16 + 0.064 + \dots$

4. Evaluate the following series or determine that they diverge (with very brief explanation).

(a) §11.2: 32

(b) §11.2: 36

(c) §11.2: 38

(d) §11.2: 44

(e) $\sum_{n=3}^{\infty} (5^{-n} + 2 \cdot 3^{-n})$

(f) $\sum_{n=1}^{\infty} \frac{7^{n-1}}{(-9)^n}$

$$(g) \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

$$(h) \sum_{n=1}^{\infty} (\tan^{-1}(n+2) - \tan^{-1} n)$$

HOMEWORK DAY 22 – *Integral Test, p -series (S 11.3)*

5. Use the Integral Test to determine whether the series is convergent or divergent. Carefully show all details.

(a) §11.3: 5

(b) §11.3: 9

(c) §11.3: 10

6. Determine whether the following series converge or diverge (with a brief, but complete, explanation).

(a) $\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt{2}}}$

(b) $1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \frac{1}{125} + \dots$

(c) $\frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \frac{1}{5\sqrt{5}} + \dots$

(d) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n}}$

(e) $\sum_{n=1}^{\infty} \frac{\sqrt{n} + 4}{n^2}$

$$(f) \sum_{n=1}^{\infty} \frac{1}{n^5}$$

$$(g) 1 + \frac{10}{4} + \frac{10}{9} + \frac{10}{16} + \dots$$

$$(h) \sum_{n=1}^{\infty} \tan^{-1} n$$

$$(i) \sum_{n=100}^{\infty} \frac{4^n}{3^n + 3}$$

$$(j) \sum_{n=2}^{\infty} 2^n$$

$$(k) \sum_{n=1}^{\infty} e^{-n}$$

HOMEWORK DAY 23 – *Comparison Tests §11.4*

7. Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 4}$ converges or diverges, using the following 3 methods. Give all necessary details in each case.

(a) Direct Comparison Test:

(b) Limit Comparison Test:

(c) Integral Test:

8. Determine whether the following series converge or diverge, using the method of your choice (with brief but complete explanation).

(a) $\sum_{n=1}^{\infty} \frac{n-1}{n^3+1}$

(b) $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}-1}$

(c) $\sum_{n=2}^{\infty} \ln n$

(d) $\frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \frac{1}{11} + \dots$

(e) $\sum_{n=1}^{\infty} \frac{1 + \cos n}{e^n}$

(f) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{3n^4 + 1}}$

$$(g) \sum_{n=1}^{\infty} \frac{\sqrt{1+n}}{2+n}$$

$$(h) \sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

$$(i) \sum_{n=1}^{\infty} (1 + \frac{1}{n})^n$$

$$(j) \sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^2 + 1}$$